DATA SCIENCE INTERNSHIP

To perform end-to-end data processing using Python and create an interactive business intelligence dashboard using Power BI.

Phase 1: Data Preparation & Cleaning (Python)

Task: Using the Python pandas library, perform the following data manipulation steps to prepare the dataset for visualization.

1. **Data Cleaning:** Load the initial dataset and check for missing or null values.
    - Standardize column headers and remove any duplicate records.
2. **Type Conversion:** Import the cleaned data.
    - Convert the **Revenue** column into an **Integer** type (ensure any currency symbols or commas are removed before conversion).
3. Create a new calculated column titled **Total Workers** by summing the values of Workers and Previous Workers.
4. **File Export:** Save the final processed dataframe as a new file named NewCompany2.csv.

Phase 2: Data Visualization (Power BI)

Task: Import the NewCompany2.csv file into Power BI and develop a single-page dashboard containing the following visual elements:

1. **City-wise Revenue:** Generate a chart showing the total revenue distribution across different cities.
2. **Metro-wise Growth:** Visualize growth trends based on Metropolitan areas.
3. **Temporal Filter:** Implement a **Button Slicer** using the Founded Year column to allow users to filter the entire dashboard by the company's age.
4. **Workforce Analysis:** Create a visualization showing the **Average number of workers** per **Industry**, utilizing the Total Workers column created in Phase 1.

Phase 3: Reporting

Task: Finalize the project by exporting the completed Power BI dashboard as a **PDF file** for stakeholder review.

Deliverables - Github:

- Python script (.py)
- The processed dataset (NewCompany2.csv)
- The Power BI report file (.pbix)
- The exported PDF dashboard.